

# A new mixed integer linear programming model for the multi level uncapacitated facility location problem <sup>☆</sup>

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## ABSTRACT

This paper considers the multi level uncapacitated facility location problem (MLUFLP). A new mixed integer linear programming (MILP) formulation is presented and validity of this formulation is given. Experimental results are performed on instances known from literature. The results achieved by CPLEX and Gurobi solvers, based on the proposed MILP formulation, are compared to the results obtained by the same solvers on the already known formulations. The results show that CPLEX and Gurobi can optimally solve all small and medium sized instances and even some large-scale instances using the new formulation.

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## 1. Introduction

The interest in solving facility location problems has been growing steadily in the past few decades. Review of a large variety of facility location problems can be found in [1]. The main focus of the research in this area has been concentrated on the location problems, which require minimization of total travel time, physical distance, or some other related cost. Very often, it is assumed that facilities have sufficiently large capacities to meet even the largest demands. Such problems are mostly called uncapacitated, in contrast to those where capacities are limited, and quite a few deterministic models are proposed for solving such problems. On the other hand, there are some models which include uncertainty [2], but describing all such approaches is out of this paper scope.

In the classical single-level uncapacitated facility location problem, the key decision is to determine the location of facilities and to assign each client to an open facility, in a way that minimizes the total cost of opening the facilities and servicing the clients. In the multilevel uncapacitated facility location problem (MLUFLP), each client must be serviced, in general, by a sequence of, at most,  $k$  different facilities. The multi level uncapacitated facility location problem is NP-hard, since it is a generalization of the (single level) uncapacitated facility location problem whose NP-hardness is proven in [3].

This paper considers a hierarchical system of facilities composed of  $k$  levels in which the lowest level of facilities is called level 1, whereas the highest level is level  $k$ . The clients can be defined as level 0. The existence of a network whose vertices represent facilities and clients is assumed. Classification of such a hierarchical system of facilities and their associated client and respective location problems is given in [4] where it was determined according to four attributes. The first attribute, flow pattern, describes the flow of services or goods on edges between vertices of the network. The other two attributes, service

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availability and spatial configuration of services, express the vertical interaction between levels of the hierarchy. The last attribute is the objective for locating the facilities.

- *Flow pattern.* It can be either single-flow or multi-flow. Single-flow starts at level 0, passes through all levels, and ends at the highest level (or it starts from the highest level and ends at level 0). It means that facilities included in flow must belong to consecutive levels. Multi-flow can be between facilities on any level, not only consecutive ones.
- *Service varieties.* There are two possibilities, nested or non-nested, depending on the service availability at various levels. In a nested case, a higher-level facility provides all the services provided by a lower level facility and at least one additional service. In a non-nested case, facilities at each level offer different services, as can be seen in [5].
- *Spatial configuration:* Depending on the spatial configuration of levels the system can be coherent or non-coherent. In the former case, all clients assigned to a particular lower-level facility are assigned to one and the same higher-level facility as in [6]. Non-coherent systems are less constrained on the spatial configuration of levels.
- *Objective:* There are three well-known types of objectives to locate facilities: median, covering, and fixed charge objectives. The median models try to minimize transportation costs between clients and facilities. In covering models, a client is considered covered by a facility if the facility is located within a specified proximity. Consequently, covering models essentially try to minimize the number of facilities needed for coverage of all clients (set covering) and to maximize covered clients with a particular number of facilities (maximum covering). The fixed charge location models try to minimize the total facility set up and transportation costs. Most of these problem types can be extended to a multiple objective optimization.

Basic applications of multilevel facility location problem (uncapacitated and capacitated) can be found, for example, in health-care systems [7,8], solid waste management systems [9], production–distribution systems [10,11], education systems [12], emergency medical service (EMS) systems [13], telecommunication networks [14,15], etc. Theoretical applications on the bounded depth Steiner tree and range assignment problems with corresponding model can be found in [16].

There are two different approaches in formulating MILP models for hierarchical location problems. Both approaches are well represented in literature: flow-based and (path) assignment-based formulations. In flow-based formulations, demand flows from facilities at one level to the next level of the hierarchy, similarly to network flow problems. The assignment-based formulations allocate demand to facilities at each hierarchical level, as it may be the case in assignment problems. Both formulations are usually used in solving minimax location problems: p-Median and fixed charge problems. The models that consider the covering location problems are out of the scope of this paper and can be found in [4].

Both flow-based and assignment-based formulations have produced several MILP models. Some of the flow-based formulation models can be found in [6,17–19] and certain assignment-based formulation models can be found in [8,17,20]. Models for both formulations can be found in [21]. The difference between the two formulations is in the way that the demand of each facility is being treated. In the flow-based formulation, the quantity of goods or services that flows through the facility, is represented by a variable, which determines the flow from one-level of the hierarchy to another. Assignment-based formulation, on the other hand, determines demand of a client by assigning a set of facilities, one from each level, which will satisfy it.

The model proposed in this paper belongs to the class of assignment-based formulations, so those models will be discussed in more detail.

## 2. Assignment-based formulations

There is a general model for assignment-based formulations of single flow multi level facility location problem. A multi flow variant can be found, for example, in [8]. Here, it is represented in the same way as in [4]. For the purpose of simplicity it will be given for the two-level case, which is the most explored in literature.

Let  $u_{ijk}$  be 1 if client  $k$  is assigned to level 2 facility  $i$  via level 1 facility  $j$ ; 0, otherwise. Let  $c_{ijk}$  is the cost charged for client  $k$  being assigned to a pair of level 1 and 2 facilities at sites  $j$  and  $i$ , respectively. Also, we define  $I$  as the set of candidate sites for level 2 facilities, and  $K$  as the set of clients. Number  $p$  represents the number of level 1 facilities to be opened. Let  $y_j$  be 1 if level 1 facility located at  $j$ , zero otherwise. Costs of opening facility  $j$  on level 1 are given with  $f_j$ . Accordingly,  $J$  is the set of candidate sites of level 1 facilities, with  $q$  as a number of facilities on level 2 which can be opened, and  $(x_i) = 1$  for all open facilities at level 2. Costs of opening those facilities are given by  $g_i$ . Now, the formulation of the two-level system, for the fixed charge objective function (more common for assignment-based formulation) is:

$$\min \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} u_{ijk} c_{ijk} + \sum_i g_i x_i + \sum_j f_j y_j \quad (1)$$

subject to:

$$\sum_{j \in J} y_j = p, \quad (2)$$

$$\sum_{i \in I} x_i = q, \quad (3)$$

$$\sum_{i \in I} \sum_{j \in J} u_{ijk} = 1, \quad k \in K, \quad (4)$$

$$u_{ijk} \leq y_j, \quad i \in I, j \in J, k \in K, \quad (5)$$

$$u_{ijk} \leq x_i, \quad i \in I, j \in J, k \in K, \quad (6)$$

$$u_{ijk} \geq 0, \quad i \in I, j \in J, k \in K, \quad (7)$$

$$x_i, y_j \in \{0, 1\}, \quad i \in I, j \in J. \quad (8)$$

The objective function (1), presented in [20], minimizes the total transportation costs from clients to level 1 facilities, and between facilities at level 1 and 2. This is done in addition to fixed costs of establishing facilities on level 1 and 2. Constraints (2) and (3) guarantee that the number of opened facilities is exactly  $p(q)$  on level 1 (2). Constraints (4) ensure that each demand is assigned exactly to one pair of level 1 and 2 facilities, while constraints (5) and (6) allow the assignment to occur only through open facilities on level 1 and 2, respectively.

Other approaches differ in certain aspects to one another. In [22], fixed costs are given for linked pairs of facilities from both levels. Paper [23] gives a generalized version of a single model for two-level systems. Approach in [24] considers the same objective function as [22] for a two-level location problem. However, in [24] this is done by capacitating lower-level facilities and only one higher-level facility, which serves the lower-level facilities. Additionally, each client demand is served by only one lower-level facility.

Approaches for solving facility location problems, found in literature, are often in a form of LP relaxation and branch-and-bound procedures with branching rules specifically devised for certain problems. In [25] the dual ascent procedure was used to solve the Lagrangean dual. It was combined with a primal descent heuristic and node simplification rules in order to be used in their proposed branch-and-bound algorithm. Algorithm was modified to solve, not only two level, but multi level case. The correctness of this approach with regard to the node simplification procedure is discussed in [26]. Rules for node simplification procedure are proposed in [20] and a greedy heuristic was applied to obtain upper and lower bounds. Another application of primal and dual procedures to the result of branch-and-bound technique is proposed in order to find optimal solutions in [22].

Paper [10] presents another approach that applies polyhedral theory on facet for single-level problem and adapts it to the two-level problem with addition of new facets. Their conclusion is that the strengthened assignment-based formulation produces tighter LP relaxation. The extension of this idea for two level case can be found in [27].

Another branch of research, especially for the two-level uncapacitated problem with fixed costs, was a development of approximation algorithms. An approximation algorithm produces a solution which quality is within some factor of the optimal solution in a reasonable time.

If the factor is  $r$ , the algorithm is called  $r$ -approximation algorithm and its solution is called  $r$ -approximation. As before, majority of these algorithms are based on LP relaxations. One of the first results for two level problem is a 4-approximation algorithm that can be found in [28]. Adaptation of this algorithm on  $k$  level case can be found in [29]. The first result for multi-level case is in [30] where a 3-approximation algorithm was developed.

The maximization version of the two level UFLP was also investigated. The approximation factor for the maximization version of the two level uncapacitated facility location problem was shown to be 0.47 in [31]. A simpler approach that obtained a worse factor of 0.5 can be found in [32].

The other direction in literature is devoted to the approximation algorithms without the use of LP relaxation. The first result obtained the factor of 9.2 and can be found in [33,34]. Further improvement led to the factor 6 in [35]. Later, the constant was improved to 4.56 in [36]. The algorithm is based on the reduction of a multi-level problem to a single-level problem. In their further research [37], the factor was decreased from 4.56 to 3.27. The same paper provided 2.42 and 2.85 approximation for two-level and three-level problems, respectively. Although all these results are based on polynomial time algorithms, practical result in reasonable times was not presented on any test instances. Newer improvements can be found in [38]. Lower bound on approximability factor and survey of other best achieved factors are presented in [39].

Because of its hardness, various authors applied heuristics in order to solve it. Some of those, especially ones which deal with multi level (2 and more) facility location problems can be found in [40,41].

This paper introduces a new model for MLUFLP with given partition of facilities presented in [42], which will be discussed further in the following section.

### 3. Previous work

Let us first present the problem as given in [42]. Four methods for solving MLUFLP were implemented. Three algorithms are based on linear programming relaxation of the model (it will be described in the following subsection), which has an enormous number of variables and a moderate number of constraints. Consequently, these three algorithms are capable to solve only small size MLUFLP instances with up to 52 potential facility locations. The running times of these methods are greater than 100 s. Gap values are relatively satisfactory, but the corresponding running times are very large even for these small size instances. Obviously, all three linear programming approaches are unable to solve larger size problem instances. The fourth method produces the results very quickly (up to 0.07 s), but the gaps are very large – up to 732%.

The first improvement of problem presented in [42] was given in [37], which introduces a modified reduction with a constant factor.

These facts indicate that finding a new mathematical formulation, which can help in solving larger MLUFLP instances, is justified, also creating the opportunity for the usage of metaheuristics for quick solving the large-scale problems.

An evolutionary approach was introduced in papers [43,44]. This approach used binary encoding and standard genetic operators. Objective function evaluation was performed using dynamic programming. The experiments were carried out on modified standard single level facility location problem instances. Genetic algorithm (GA) outperformed previous methods, reaching optimal solutions for the instances of smaller dimension. Optimal solutions were obtained by total enumeration and CPLEX solver using mathematical ILP formulation from [42]. It also produced solutions for medium and large-scale instances. Later, it was shown that some of these solutions are incorrect.

An improved genetic algorithm (GA) for solving MLUFLP was presented in [45]. First improvement was obtained by better implementation of dynamic programming, which sped-up the running time of the overall GA implementation. Second improvement is hybridization of the genetic algorithm with the fast local search procedure designed specially for MLUFLP. Improved genetic algorithm reached all previously known optimal and the best solutions, but in much shorter time. Hybridization with local search improved several best-known solutions for large-scale MLUFLP instances, in cases when optimal solution was not known. Compared to the previous GA approach, the overall running time of both proposed GA methods is significantly shorter.

#### 3.1. Problem description

The input data for the MLUFLP consist of a set of facilities  $F$  ( $|F| = m$ ) partitioned into  $nl$  levels, denoted as  $F_1, \dots, F_{nl}$ , a set of clients  $D$  ( $|D| = n, F_{nl+1} = D$ ), a fixed cost  $f_i$  for establishing the facility  $i \in F$ , and a matrix of the transportation costs  $c_{ij}$  for  $i \in F_{k+1}, j \in F_k, k = 1, \dots, nl$ . A feasible solution assigns each client a sequence of  $nl$  established facilities, one from each level  $F_{nl}, \dots, F_1$ , respectively. A feasible solution is charged the sum of the fixed costs of the facilities used, plus the transportation costs of the clients' assignments. Each client's transportation cost is the sum of the transportation cost from itself to the first facility of its sequence, plus the transportation cost between successive facilities in the sequence. An optimal solution is a feasible solution with a minimum total cost.

#### 3.2. An ILP formulation

This section presents the integer linear programming formulation of the MLUFLP problem presented in [42], in following text called EDW model. The assignment of a client  $j \in D$  to a valid sequence of facilities is represented as the assignment of  $j$  to a path  $p$  from  $j$  to one of the top-level facilities  $F_1$ . The set of all valid sequences of facilities is defined by  $P = F_k \times \dots \times F_1$  and the transportation cost of the client  $j$ 's assignment to sequence  $p = (i_k, \dots, i_1)$  by  $c_{pj} = c_{ji_k} + c_{i_k i_{k-1}} + \dots + c_{i_2 i_1}$ . Variable  $y_i$  has value 1 if the facility  $i$  is established, 0 otherwise. Similarly, variable  $x_{pj}$  represents whether or not a client  $j$  is assigned to the path  $p$ . Using the notation mentioned above, the problem can be written as:

$$\min \sum_{i \in F} f_i y_i + \sum_{p \in P} \sum_{j \in D} c_{pj} x_{pj}, \quad (9)$$

$$\sum_{p \in P} x_{pj} = 1, \quad j \in D, \quad (10)$$

$$\sum_{p \ni i} x_{pj} \leq y_i, \quad i \in F, j \in D, \quad (11)$$

$$x_{pj} \in \{0, 1\}, \quad p \in P, j \in D, \quad (12)$$

$$y_i \in \{0, 1\}, \quad i \in F. \quad (13)$$

The objective function (1) minimizes the sum of overall transportation costs and fixed costs for establishing the facilities. The constraint (2) ensures that each client is assigned to a path, while the constraint (3) guarantees that any facility on a path

used by some client is established. Constraints (4) and (5) reflect the binary nature of variables  $x_{pj}$  and  $y_j$ . Number of  $x$  variables is equal to  $|D| \cdot \prod_{k=1}^{nl} |F_k|$ , where number of  $y$  variables is equal to the number of facilities. It is obvious that the number of binary variables can be very large. The problem is illustrated with the following example.

**Example 1.** Let us assume that there are  $k = 2$  levels of  $m = 6$  facilities: the first level contains 2 potential facilities and the second level contains 4 potential facilities. In this example, there are  $n = 5$  clients to be served. Fixed costs of establishing the facilities are given in Table 1 and the distances between facilities of different levels and the distances between clients and facilities on the second level are given in Tables 2 and 3, respectively. Graphical representation of the problem can be seen in Fig. 1.

The total enumeration technique, is used to obtain an optimal solution. Established facilities are: Fac2 on the first and Fac3, Fac5 on the second level. The objective function value is 329. The sequences of facilities for each client are shown in Table 4. Graphical representation of optimal solution can be seen in Fig. 2, black squares representing not established facilities.

### 3.3. Other ILP formulations

This section presents the majority of models for assignment-based formulation with fixed costs for multi level (3 or more) facility location problem as defined in [4] and given comparison to the EDW model.

In [26] problem definition and ILP formulation is given in the following way. Let  $N_r$  for  $r = 1, \dots, p$  be the set of sites where facilities of level  $r$  can be selected. To each facility  $j_r$  is associated a fixed cost  $f_{j_r}$ . Let us denote the set of clients by  $I = 1, \dots, m$  and the profit obtained by serving client  $i$  from the  $p$  facilities  $j_1, \dots, j_p$  by  $c_{ij_1 \dots j_p}$ . For each given subset  $S_r \subset N_r$ ,  $r = 1, \dots, p$ , of opened facilities the corresponding solution is given by assigning each client to the most profitable combination of  $p$  open facilities. From this it follows that the multi level uncapacitated facility location problem can be formulated as:

$$\max_{S \subset N} Z(S) = \sum_{i \in I} \max_{j_1 \in S_1, \dots, j_p \in S_p} c_{ij_1 \dots j_p} - \sum_{r=1}^p \sum_{j_r \in S_r} f_{j_r}, \quad (14)$$

where  $N = \bigcup_{r=1}^p N_r$  and  $S = \bigcup_{r=1}^p S_r$ . Now, this model can be reduced to EDW model if we observe that, due to fact that facilities are already partitioned, sets  $N_r$  are already defined. From matrix of transportation costs between facilities on adjacent levels and transportation costs from clients and facilities on the lowest level all coefficients  $c_{ij_1 \dots j_p}$  can be evaluated. Since in our problem we would like to minimize transportation costs, both max functions should be converted to min, and subtraction should be changed to addition. With this changes, model given in [26], can be reduced to EDW model. Also, let us note that in

**Table 1**  
Fixed costs.

| Facilities | Fac1 | Fac2 | Fac3 | Fac4 | Fac5 | Fac6 |
|------------|------|------|------|------|------|------|
| Fixed cost | 70   | 50   | 30   | 20   | 20   | 40   |

**Table 2**  
Distance between facilities on the level 2 and the level 1.

|      | Fac1 | Fac2 |
|------|------|------|
| Fac3 | 58   | 23   |
| Fac4 | 44   | 74   |
| Fac5 | 67   | 15   |
| Fac6 | 29   | 38   |

**Table 3**  
Distance between clients and facilities.

|          | Fac3 | Fac4 | Fac5 | Fac6 |
|----------|------|------|------|------|
| Client 1 | 38   | 13   | 57   | 41   |
| Client 2 | 47   | 52   | 63   | 15   |
| Client 3 | 42   | 48   | 13   | 54   |
| Client 4 | 9    | 54   | 41   | 43   |
| Client 5 | 15   | 18   | 36   | 22   |

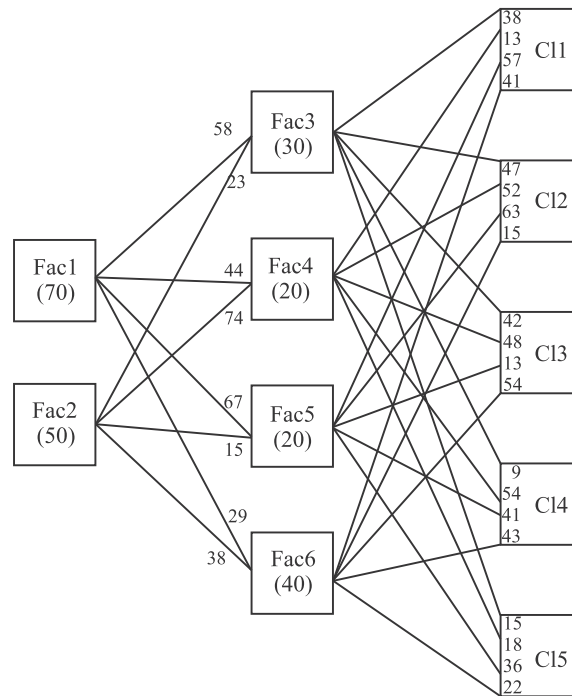


Fig. 1. Facilities and clients with fixed and transportation costs.

**Table 4**  
Sequences of facilities for clients.

|          | level 2 | level 1 |
|----------|---------|---------|
| Client 1 | Fac3    | Fac2    |
| Client 2 | Fac3    | Fac2    |
| Client 3 | Fac5    | Fac2    |
| Client 4 | Fac3    | Fac2    |
| Client 5 | Fac3    | Fac2    |

this formulation each client is served from facilities on level 1 instead those on the highest level but this can easily be solved by reversing the ordering of facilities.

Another model can be found in [30]. We will first give their presentation and afterwards discuss the similarities and differences to the EDW model.

Let  $D$  be the set of clients. Each client  $j \in D$  must be assigned to precisely one facility at each of the  $k$  levels. Let  $F^l$  be the set of sites where facilities on level  $l$ ,  $1 \leq l \leq k$ , may be located, and assume that the sets  $F^l$  are pairwise disjoint. We denote the set of all such sites,  $\bigcup_{l=1}^k F^l$ , by  $F$ . The cost of setting up a facility at site  $i$  is  $f_i$ . We assume that  $f_i > 0$  for each  $i \in F$ . The cost of shipping between points  $i, j \in F \cup D$  is equal to  $c_{ij}$ . The following assumptions on the service costs are valid:

- $c_{ij} > 0$ , for each  $i, j \in F \cup D$ ;
- $c_{ij} = c_{ji}$ , for each  $i, j \in F \cup D$ , i.e., the service costs are symmetric;
- $c_{ik} \leq c_{ij} + c_{jk}$ , for each  $i, j, k \in F \cup D$ , i.e., the service costs satisfy the triangle inequality.

The symbol  $p$  will be used to denote a sequence of facilities  $i_l \in F^l$ ,  $l = 1, \dots, k$ , and shall refer to  $p$  as a path of facilities. The set of all possible paths is denoted by  $P$ . Each client must be assigned to precisely one path  $p \in P$ . The total service cost incurred by assigning client  $j$  to path  $p = (i_1, i_2, \dots, i_k)$  is equal to  $c_{pj} = c_{i_1, i_2} + c_{i_2, i_3} + \dots + c_{i_{k-1}, i_k} + c_{i_k, j}$ .

Let  $x_{pj}$  be equal to 1 if client  $j$  is assigned to path  $p$ , and 0 otherwise. Furthermore, let  $y_{i_l}$  be equal to 1 if facility  $i_l$  is open at level  $l$ . Now, formulation can be stated as:

$$z = \min \sum_{l=1}^k \sum_{i_l \in F^l} f_{i_l} y_{i_l} + \sum_{p \in P} \sum_{j \in D} c_{pj} x_{pj} \quad (15)$$

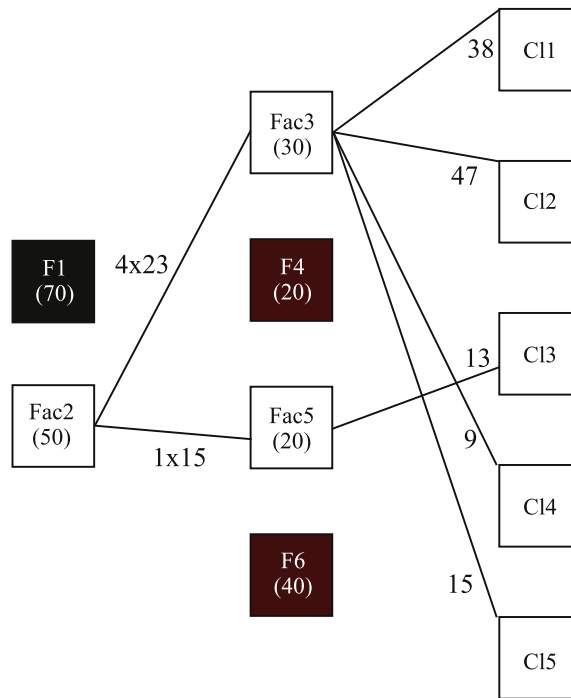


Fig. 2. Optimal solution for the Example 1.

subject to:

$$\sum_{p \in P} x_{pj} = 1, \quad j \in D, \quad (16)$$

$$\sum_{p: i_l \in p} x_{pj} - y_{i_l} \leq 0, \quad j \in D, i_l \in F^l, l = 1, \dots, k, \quad (17)$$

$$x_{pj} \geq 0, \quad p \in P, j \in D, \quad (18)$$

$$x_{pj} \in \{0, 1\}, \quad p \in P, j \in D, \quad (19)$$

$$y_{i_l} \in \{0, 1\}, \quad i_l \in F^l, l = 1, \dots, k. \quad (20)$$

In problem description from 3.1, facilities are already partitioned in levels, instead of finding their assignment as in formulations above. However, if corresponding  $y$  variables are given values 0 or 1 (so that assignment is equal to apriori partitioning) and if corresponding transportation costs  $c$  are defined as transportation costs from problem description in 3.1 and there exists a link between sites  $i$  and  $j$  and infinity or a very large number otherwise, the formulation from [30] is reduced to EDW model.

Approaches very similar in ideas of interpreting the MLUFLP to the previous model can be found in [37,46] and will not be discussed in detail. Paper [46] gives yet another survey of models for MLUFLP and all related problems with theoretical discussion on their quality.

Another formulation with a different approach is presented in [47], denoted as the GAB model in further text. The main idea is that, instead of assigning clients to paths, the clients are assigned to edges, such that each client is assigned to an edge between each two consecutive levels of facilities and the edges have a vertex in common. For purposes of modeling, the authors introduced the following 0–1 variables:

- $y_i$ ,  $i \in F$  indicate whether  $i \in F$  is open;
- $x_{ki}$ ,  $i \in F_1, k \in D$ , indicate whether client  $k$  is assigned to facility  $i \in F_1$ ;
- $z_{kij}$ ,  $(i, j) \in F_l \times F_{l+1}$ , for  $l = 1, \dots, n-1$  indicate whether client  $k$  uses the edge  $(i, j)$ .

The transportation costs is denoted by

$$c(x, z) := \sum_{k \in D} \sum_{i \in F_1} c_{ki} x_{ki} + \sum_{k \in D} \sum_{l=1}^{n-1} \sum_{(i,j) \in F_l \times F_{l+1}} c_{ij} z_{kij} \quad (21)$$

and the costs for opening facilities by:

$$f(y) := \sum_{i \in F} f_i y_i. \quad (22)$$

Now, problem can be formulated as:

$$\min \quad c(x, z) + f(y) \quad (23)$$

subject to:

$$\sum_{i \in F_1} x_{ki} = 1, \quad k \in D, \quad (24)$$

$$\sum_{j \in F_2} z_{kij} = x_{ki}, \quad i \in F_1, \quad k \in D, \quad (25)$$

$$\sum_{j \in F_{l+1}} z_{kij} = \sum_{j' \in F_{l-1}} z_{kj'i}, \quad i \in F_1, \quad 2 \leq l \leq n-1, \quad k \in D, \quad (26)$$

$$x_{ki} \leq y_i, \quad k \in D, \quad i \in F_1, \quad (27)$$

$$\sum_{j \in F_{l-1}} z_{kij} \leq y_i, \quad 2 \leq l \leq n, \quad i \in F_1, \quad k \in D, \quad (28)$$

$$y_i \in \{0, 1\}, \quad i \in F, \quad (29)$$

$$x_{ki} \in \{0, 1\}, \quad k \in D, \quad i \in F_1, \quad (30)$$

$$z_{kij} \in \{0, 1\}, \quad (i, j) \in V_l \times V_{l+1}, \quad 1 \leq l \leq n-1, \quad k \in D. \quad (31)$$

Constraints (24) ensure that each client  $k \in D$  gets connected to exactly one facility on the first level. Constraints (25) enable that client  $k$  uses an edge  $(i, j) \in F_1 \times F_2$  only if  $k$  is assigned to facility  $i \in F_1$ , i.e.,  $x_{ki} = 1$ . Constraints (26) ensure that client  $k$  uses an edge  $(i, j) \in F_l \times F_{l+1}$ ,  $2 \leq l \leq n-1$  only if  $k$  uses an edge  $(j', i)$ , with  $j' \in F_{l-1}$ . Finally, constraints (27) and (28) provide that a client  $k$  will be assigned to a facility  $i \in V_1$  and will use an edge  $(j, i) \in F_{l-1} \times F_l$ , for  $2 \leq l \leq n-1$ , only if facility  $i$  is open.

Here, as in the first model of this section, clients are served from facilities on level 1 instead of the ones on the highest level, as in the EDW model. However, as we remarked earlier, it can easily be solved by reversing the ordering of levels. This model cannot be easily reduced on the EDW model, so it will be tested separately in Section 5 by both solvers, CPLEX and Gurobi.

#### 4. A new mixed integer linear programming formulation

Now, the new mixed integer linear programming (MILP) formulation will be proposed.

Let variable  $y_i$  has the value 1 if the facility  $i$  is established and 0 otherwise. In fact, the  $y$  variables have the same role as in ILP model (EDW model). Variable  $z_{ik}$  can be considered as the quantity of goods that facility  $i$  gets through facility  $k$  from previous level.

The new MILP model of the MLUFLP can now be formulated as:

$$\min \sum_{i \in F} f_i y_i + \sum_{l=1}^{nl} \sum_{i \in F_{l+1}} \sum_{k \in F_l} c_{ik} z_{ik} \quad (32)$$

subject to:

$$\sum_{k \in F_{nl}} z_{ik} = 1, \quad i \in D, \quad (33)$$

$$\sum_{k \in F_{l-1}} z_{ik} = \sum_{q \in F_{l+1}} z_{qi}, \quad l = 2, \dots, nl; \quad i \in F_l, \quad (34)$$

$$z_{ik} \leq n \cdot y_k, \quad i \in F_{l+1}, \quad k \in F_l, \quad l = 1, \dots, nl, \quad (35)$$

$$z_{ik} \in \mathbb{R}^+ \cup \{0\}, \quad (36)$$

$$y_i \in \{0, 1\}, \quad i \in F. \quad (37)$$

This new formulation has much less decision variables than the EDW model. Number of  $z$  variables is equal to  $\sum_{k=1}^{nl} |F_k| \cdot |F_{k+1}|$  which is usually much less than  $\prod_{k=1}^{nl+1} |F_k|$ . Number of  $y$  variables is the same because they have the same role in both models. This is also true concerning the comparison of decision variables for GAB model and the new formulation. Let us now prove that this new formulation is equivalent to the EDW model. This will be done through three lemmas.

**Lemma 1.** *For every feasible solution of the EDW model, there is a feasible solution of the new MILP model, with the same value of the objective function.*

**Proof.** Let  $x_{pj}$ ,  $p \in P$ ,  $j \in D$  and  $y_i$ ,  $i \in F$  be a feasible solution of the EDW model. Established facilities are exactly those  $i \in F$  for which  $y_i = 1$ . This set is not empty. Otherwise, from (11), all the variables  $x_{pj} = 0$ , contrary to the condition (10). Because of (10), exactly one path  $p \in P$  such that  $x_{pj} = 1$  corresponds to each client  $j$ , so the feasible solution of the EDW model corresponds to  $n$  paths, leading from each client over established facilities in decreasing order of levels from  $F_{nl}$  to  $F_1$ . Let us now define the respective feasible solution of the new model. For every  $\alpha \in F_{k+1}$ ,  $\beta \in F_k$ ,  $k = 1, 2, \dots, nl$ , let us define variable  $z_{\alpha\beta}$  as the total number of paths which pass through  $\alpha$  and  $\beta$ . Let the values of  $y$  variables be the same as in the EDW model.

We now prove that conditions (33)–(37) are satisfied and that value of the objective function (32) is the same as the value reached in the EDW model. Conditions (36) and (37) are satisfied from the definition of the variables  $y$  and  $z$ .

Because of the condition (10), there is only one path from each client, so there is only one established facility on the last level on that particular path. Therefore, it is obvious that the condition (33) is satisfied.

In (34), left side of the condition is equal to the number of established paths' entries in non-terminal facility  $i \in F_l$  and right side is equal to number of exits from it. Those two numbers are equal, because every path that enters such facility must exit from it.

If  $y_k = 0$  then the facility  $k$  is not established, so there is no path through it, which implies that  $z_{ik} = 0$  for every  $i$  on the next level. In the case where  $y_k = 1$ , the number of paths through that facility cannot be greater than the number of clients ( $n$ ). Therefore, the conditions (35) are satisfied.

Both objective functions contain  $\sum_{i \in F} y_i$  which has the same value in both models. The remaining part of the objective function in the EDW model is equal to the costs of all paths from each client, i.e.  $\sum_{p \in P} \sum_{j \in D} c_{pj} x_{pj}$ . Each path contains exactly one part between established facilities on the subsequent levels. Let us denote them as  $i \in F_{l+1}$ ,  $k \in F_l$ . In the objective function of the EDW model, the cost  $c_{ik}$  appears exactly  $z_{ik}$  times. Therefore, this value is equal to  $\sum_{l=1}^{nl} \sum_{i \in F_{l+1}} \sum_{k \in F_l} c_{ik} z_{ik}$ .  $\square$

**Lemma 2.** *The new MILP model (32)–(37) has an optimal integer solution.*

**Proof.** As it is proven in the Lemma 1, from any feasible solution of the existing problem a feasible solution of the new MILP model can be constructed. Consequently, solution set of the new MILP model is non-empty. Set of feasible solutions is also bounded because all variables are non-negative and are not greater than  $n$ . Now, optimal solution of new MILP model is guaranteed by Weierstrass's theorem. Let us now prove that integer optimal solutions exist. Let us consider a feasible solution  $z_{\alpha\beta}$ ,  $\alpha \in F_{k+1}$ ,  $\beta \in F_k$ ,  $k = 1, 2, \dots, nl$  and  $y_i$ ,  $i \in F$ .

**Table 5**  
Experimental results of CPLEX and Gurobi solvers.

| Instance name           | Optimal solution | EDW model |           | GAB model |           | New MILP model |           |
|-------------------------|------------------|-----------|-----------|-----------|-----------|----------------|-----------|
|                         |                  | $t_{CPL}$ | $t_{Gur}$ | $t_{CPL}$ | $t_{Gur}$ | $t_{CPL}$      | $t_{Gur}$ |
| cap71_2L_6_10.50        | 1813375.512500   | 0.03      | 0.015     | 0.062     | 0.016     | 0.031          | 0.015     |
| cap71_3L_2_5_9.50       | 4703216.306250   | 0.11      | 0.031     | 0.109     | 0.031     | 0.031          | 0.015     |
| cap101_2L_8_17.50       | 1581551.393750   | 0.12      | 0.046     | 0.171     | 0.046     | 0.093          | 0.031     |
| cap101_3L_3_7_15.50     | 3227179.812500   | 0.23      | 0.093     | 0.202     | 0.046     | 0.062          | 0.015     |
| cap131_2L_13_37.50      | 1592548.450000   | 0.33      | 0.141     | 0.608     | 0.140     | 0.124          | 0.093     |
| cap131_3L_6_14_30.50    | 3201970.462500   | 2.20      | 0.982     | 0.843     | 0.218     | 0.280          | 0.125     |
| cap131_4L_3_7_15_25.50  | 3630297.668750   | 9.64      | 4.118     | 0.811     | 0.187     | 0.265          | 0.078     |
| capa_2L_30_70.1000      | 14829245.628060  | –         | 21.9      | –         | 19.06     | 23.53          | 10.61     |
| capa_3L_15_30_55.1000   | 25424361.913455  | –         | –         | –         | 30.01     | 43.52          | 10.50     |
| capa_4L_6_12_24_58.1000 | 35421258.152110  | –         | –         | –         | 33.37     | 16.91          | 17.46     |
| capb_2L_35_65.1000      | 14479223.791075  | –         | 23.24     | –         | 18.97     | 39.87          | 21.89     |
| capb_3L_12_25_63.1000   | 25986997.287390  | –         | –         | –         | 28.63     | 35.24          | 21.26     |
| capb_4L_6_13_31_50.1000 | 41787432.238230  | –         | –         | –         | 37.94     | 47.97          | 17.86     |
| capc_2L_32_68.1000      | 14072575.517910  | –         | 22.17     | –         | 14.29     | 32.57          | 13.92     |
| capc_3L_13_27_60.1000   | 26751918.745245  | –         | –         | –         | 23.12     | 46.83          | 21.42     |
| capc_4L_4_9_27_60.1000  | 47109818.658030  | –         | –         | –         | 31.47     | 84.22          | 23.42     |
| mq1_2L_100_200.300      | 8341.287         | –         | –         | –         | –         | 29576          | 191135    |
| mq1_3L_30_80_190.300    | 12994.871        | –         | –         | –         | –         | 35241          | 73428     |
| mq1_4L_18_39_81_162.300 | 18048.0305       | –         | –         | –         | –         | 53579          | 148718    |
| mq1_4L_20_40_80_160.300 | 17648.0095       | –         | –         | –         | –         | 40694          | 53206     |

Firstly, let us prove that on every level  $l = 1, 2, \dots, nl$ , there exists at least one variable  $y_k$ ,  $k \in F_l$  which has the value equal to 1. This means that at least one facility is established on each level. Contrary, suppose that on some level  $F_l$ ,  $l \in \{1, 2, \dots, nl\}$ , all variables  $y_k$  for each  $k$  are equal to 0. According to (35), all the values of the variables  $z_{ik}$  for each  $i \in F_{l+1}$ ,  $k \in F_l$  would also equal 0. Then, the right side in condition (34) would be equal to 0 and so would be the left side. Consequently, all the variables  $z_{ik}$ , for  $i \in F_l$ ,  $k \in F_{l-1}$  would equal 0. Repeating this reasoning would result that  $z_{ik} = 0$  whenever  $i \in F_l$ ,  $k \in F_{l-1}$  for all levels  $l, l+1, \dots, nl$ . But, the last conclusion is contrary to (33). Consequently, the statement is proven.

For any chosen (fixed) values of the variables  $y_i$ , which are not all equal to 0 on any single level, the original problem can be seen as the transshipment problem. This corresponds to the problem for transporting a single quantity of the goods from the nodes (in this case clients) over the transfer nodes (in this case established facilities) ordered on the levels  $nl, nl-1, \dots, 2, 1$ . This transshipment problem has an optimal integer solution because the matrix generated by its conditions is totally unimodular [48]. This concludes the proof of the Lemma 2.  $\square$

**Lemma 3.** Let  $z_{\alpha\beta}$ ,  $\alpha \in F_{k+1}$ ,  $\beta \in F_k$ ,  $k = 1, 2, \dots, nl$  and  $y_i \in F$  be an integer feasible solution of the new MILP model. It corresponds to some solution of the EDW model with the same value of objective function.

**Proof.** Let us consider the graph corresponding to the transshipment problem constructed in Lemma 2, obtained from the values of the variables  $y_i$ . The edges correspond to the variables  $z_{ik} \neq 0$  for  $i \in F_{l+1}$ ,  $k \in F_l$ ,  $l = 1, 2, \dots, nl$ . Let us assign the costs  $c_{ik}$  to these edges. Because of the condition (33), a single quantity of goods from exactly one facility on the level  $nl$  is transported to every client  $i$ . To this facility, integer quantity of goods is transported from some facility on level  $nl-1$ . This process is repeated until level 1 is reached. It means that there is a path from every client through established facilities on each level starting from the first to the last level, and the whole transport from starting facilities to the clients is accomplished. Based on these paths, binary variables  $x_{pi}$  from the EDW model are constructed. Variables  $y_i$  have the same values (and meanings) in both models. Equality of the values of the objective functions from both models is given in the proof of Lemma 1 and holds in both directions. Therefore, the objective function value of the constructed feasible solution (corresponding to the EDW model) is equal to the objective function value of the new model.  $\square$

These three lemmas prove the equivalence of the EDW model and new model. In the following section, some experimental results which demonstrate superiority of the new model will be given.

Now, let us give an example to demonstrate the differences between these two models.

**Example 2.** The data used in this example will be the same as in Example 1.

The EDW model. There are two facilities on level 1 and four facilities on level 2. It means that there are 8 paths between facilities on these two levels. There are 5 clients which means that there are 40 variables  $x$ , from  $x_{1,1}$  to  $x_{8,5}$ . The total number of variables  $y$  is 6 because there are 6 facilities in total. The optimal solution obtained by CPLEX is

$$x_{5,1} = x_{5,2} = x_{5,4} = x_{5,5} = x_{7,3} = 1$$

and

$$y_2 = y_3 = y_5 = 1.$$

All other variables are equal to 0. Fixed costs for facilities Fac2, Fac3 and Fac5 are 50, 30 and 20 respectively, which in sum gives 100. Transport costs are: 23, between facilities Fac2 and Fac3 (this path is used 4 times), and 15, between facilities Fac2 and Fac5, which in total gives 109. Transport costs between the facility Fac3 and the clients C11, C12, C14 and C15 are respectively 38, 47, 9 and 15, and the transport cost between the facility Fac5 and the client C13 is 13, in total 122. So, the total sum of all costs is 329 and it is equal to the one in Example 1.

New mixed ILP model. Because there are two facilities on level 1 and four facilities on level 2, there would be 8 variables  $z$  with the indices  $z_{3,1}, z_{3,2}, z_{4,1}, \dots, z_{6,2}$ . As said, clients C1 to C5 will be treated as the facilities F7 to F11. Now, because there are 4 facilities on level 2 and 5 facilities on level 3 (these are clients) there would be 20 variables  $z$  with the indices  $z_{7,3}, z_{7,4}, z_{7,5}, z_{7,6}, z_{8,1}, \dots, z_{11,6}$ . In total, there are 28 variables  $z$ . Variables  $y$  are the same as in the EDW model and there are 6 of them. The optimal solution for this example obtained by CPLEX is:

$$z_{3,2} = 4, \quad z_{7,3} = z_{8,3} = z_{10,3} = z_{11,3} = z_{9,5} = 1$$

and

$$y_2 = y_3 = y_5 = 1.$$

All other variables are equal to 0. It is easy to calculate the value of objective function and it is equal to 329.

## 5. Experimental results

All tests were carried out on an Intel Core I3-2350M 2.3 GHz computer with 2 GB memory by single core testing option. For the testing, CPLEX 12.4 and Gurobi 5.02 solvers were used.

The models were tested on the same instances as in [43,44]. In this paper, only those instances with multiple levels of facilities are considered, since for the basic problem with only one level, all models seem to give similar results. Therefore, instances with multiple levels are the main interest of this paper, so the results on the instances with only one level of facilities are omitted.

The names of the instances are given in first column. The instance's name carries information about the number of levels, the number of facilities on each level and the number of clients, respectively. For example, the instance *capb\_4L\_6\_13\_31\_50\_1000* is created by modifying the ORLIB instance *capb*, which has 4 levels with 6, 13, 31 and 50 facilities, respectively and 1000 clients.

The second column contains the optimal solution of the current instance. In the next two columns running times  $t$  of CPLEX and Gurobi solvers on the EDW model are given. The mark “–” was used in cases when solvers could not reach an optimal solution because of “out of memory” issues. The following two columns are devoted to the GAB model. In the last two columns running times obtained by solvers using new MILP model are given.

The EDW model achieves optimal solutions in cases when the number of clients and facilities is up to 50. Additionally, Gurobi solver on EDW model optimally solved some of the medium-sized instances, but only in cases when there are only two levels of facilities. As it is indicated in Table 5, the “out of memory” issue appears in all other cases of the medium size instances when the EDW model is applied. This is a consequence of the enormous number of binary variables appearing in that model.

The GAB model has also a large number of binary variables but its structure enables both solvers to produce optimal solutions for all small size instances. CPLEX solver, due to “out of memory” issues, did not reach solutions for medium size instances. On the other hand, Gurobi obtained all optimal solutions for medium size instances in a relatively short time.

Both CPLEX and Gurobi solvers using the new model provide the optimal solutions for all small and medium-sized instances in a relatively short time. Running times of Gurobi solver are smaller, except in one case, compared to GAB and EDW models. Furthermore, CPLEX and Gurobi solvers optimally solve some of large-scale instances. For instance *mq1\_2L\_100\_200\_300* CPLEX solver using the new model produced optimal solution value 8341.287 in 29576 s. Note that, this instance is out of reach for CPLEX and Gurobi solvers applied on EDW and GAB models, since they cannot solve even the initial LP relaxation due to the enormous number of decision variables. For example, numbers of decision variables and constraints are: GAB model 6060300 binary variables and 150300 constraints, EDW model 6000300 binary variables and 90300 constraints, while the new model has only 300 binary and 80000 continuous variables with 80500 constraints. These facts clearly show the superiority of the new model over the ones from literature.

## 6. Conclusions

This paper deals with the multilevel uncapacitated facility location problem. New mixed integer linear programming formulation is proposed. Furthermore, equivalence between the ILP model (EDW model) and the new model is proven. In comparison to the EDW and GAB models, the numbers of variables for large-scale instances with more than 1 level, are significantly reduced in the new model. All models are tested by CPLEX and Gurobi solvers on instances from literature. By using the new model, medium-size instances can be optimally solved with both solvers in relatively short period of time. Furthermore, CPLEX and Gurobi solvers on the new model optimally solve some of large-scale instances which are out of reach when applied on EDW and GAB models.

This research can be extended in several ways. It would be desirable to investigate the application of some exact method using the proposed MILP formulation. Further research should also be directed at solving similar problems and testing on parallel computers.

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